A Spectrum Sensor for CubeSat Radios

Dylan J. Gormley and Anthony A. Stock
Cognitive Signal Processing Branch
NASA Glenn Research Center
Cleveland, OH, USA
dylan.j.gormley@nasa.gov; anthony.stock@nasa.gov

***Abstract*—Small-spacecraft missions increasingly use distributed architectures, making awareness of shared radio-frequency spectrum useful for interference mitigation and spectrum sharing. Conventional spectral-correlation analyzers, however, can demand more memory and computation than some CubeSat radios can readily provide. This paper develops a low-memory spectrum sensor for field-programmable gate arrays based on a streaming spectral-correlation analyzer. Under the evaluated signal model, the sensor searches configured symbol-rate and center-frequency candidates for power-of-two-order quadrature amplitude modulation waveforms with square-root-raised-cosine pulse shaping. A power-normalized binary threshold nominates symbol-rate candidates; a center-frequency detector retains the strongest frequency candidate, and a proposed scheduler coordinates parallel detector instances. Monte Carlo simulations with two co-channel signals were used to characterize the detector. Across the nine reported scenarios, the symbol-rate detection proportion ranged from 0.89 to 1.0, while the false-alarm proportion per unoccupied symbol-rate candidate ranged from 0.0013 to 0.1213. These metrics assess symbol-rate detection only: the archived scoring code does not test center-frequency correctness. Separate returned center-frequency estimates appeared more dispersed as the nominal energy-per-bit-to-noise-density setting decreased. The results support feasibility within the simulated signal conditions, but they do not establish general channel independence, hardware performance, or center-frequency accuracy.**

***Index Terms*—CubeSat, spectrum sensing, spectral correlation analyzer, field-programmable gate array**

I. Introduction

By April 2021, the cumulative number of launched nanosatellites and CubeSats had risen rapidly from the levels observed two decades earlier (Fig. 1). Distributed small-spacecraft concepts spanned different scales: HelioSwarm used a nine-spacecraft baseline [1]; the four-spacecraft Starling1 technology demonstration was intended to exercise capabilities scalable to at least 100 spacecraft [2]; and separate commercial low-Earth-orbit communications proposals contemplated thousands of satellites [3]. The last category comprises large constellations rather than CubeSat formation-flying missions. NASA Ames Research Center also illustrated several notional applications enabled by coordinated small-spacecraft swarms (Figs. 2–5) [4].

If such distributed architectures become more common, their inter-satellite links may face congestion and interference in shared bands. Spectrum sensing could provide measure-

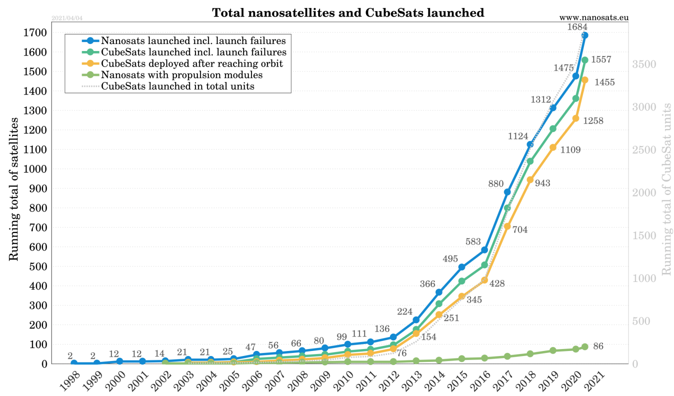

Fig. 1. Cumulative launched nanosatellites and CubeSats through April 4, 2021. Source: Erik Kulu, Nanosats Database [5].

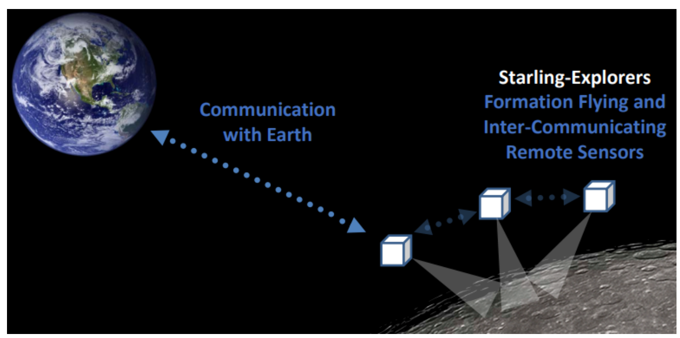

Fig. 2. Starling-Explorers concept for coordinated flight, communication, and sensors such as ground-penetrating radar or magnetometers; adapted from [4].

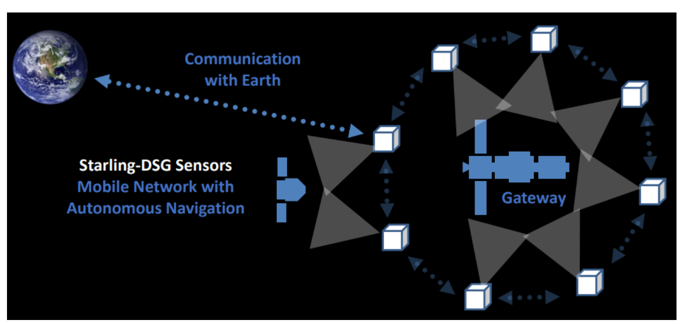

Fig. 3. Starling-DSG concept for visual inspection and monitoring in support of Deep Space Gateway operations; adapted from [4].

NASA's Cognitive Communications Project supported this work. This is a post-publication fact-checked revision dated September 2026, not the IEEE version of record.

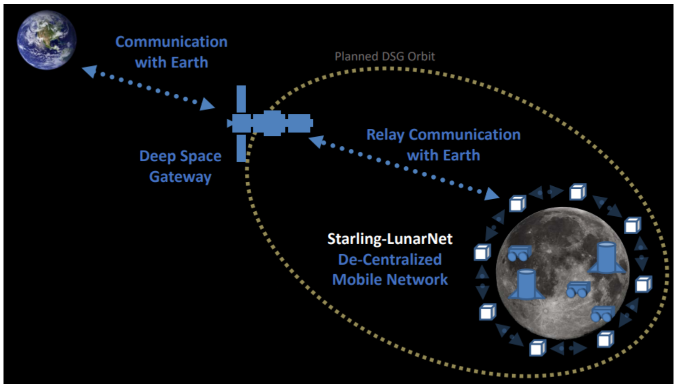

Fig. 4. Starling-LunarNet concept for monitoring the lunar surface with an ad hoc mesh network in low lunar orbit; adapted from [4].

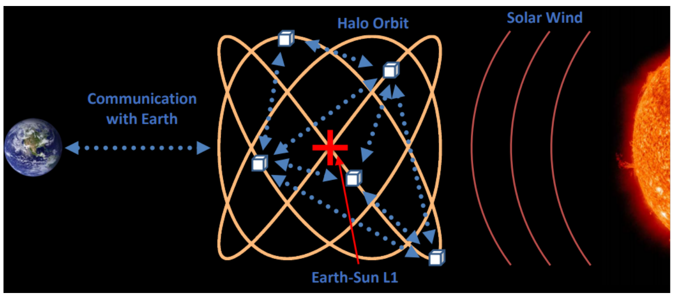

Fig. 5. Starling-SolarWind concept for autonomous radio tomography and in situ measurements from a Sun–Earth L1 halo orbit; adapted from [4].

ments to future dynamic spectrum-management algorithms, but conventional cyclostationary processing can be computationally and memory intensive [6]. CubeSat radios also operate under restrictive size, mass, power, and compute budgets. This motivates the low-memory streaming spectral correlation analyzer (streaming-SCA) developed in [7].

The streaming-SCA estimates selected points of the spectral correlation function (SCF) for square-root-raised-cosine (SRRC) pulse-shaped QAM waveforms. The evaluated implementation assumes a power-of-two modulation order, a configured search grid, and candidate symbol rates and center frequencies; it therefore is not fully blind. For requested symbol rate R_s, let

$$v = \mathrm{round}\left(\frac{R_s N}{f_s}\right)$$

be its zero-based Goertzel/DFT bin, and let $f_{c,k}$ be the kth configured physical center-frequency candidate. In zero-based sample notation, the archived estimator is

$$\widehat{S}_X^v[k] = \frac{1}{N}\sum_{n=0}^{N-1}\left|\left(X[n]e^{+j2\pi f_{c,k}n/f_s}\right) * h[n]\right|^2 e^{-j2\pi vn/N}. \tag{1}$$

Here, n is the integer sample index, N is the number of samples used for one estimate, f_s is sample rate, $*$ denotes convolution, and $h = [1/2, 1/2]$ is the two-tap low-pass filter. The positive channelizer sign matches the archived transmitter's negative carrier convention. The one-based Goertzel call uses bin $v+1$, while k indexes the configured center-frequency list. The quantity $\widehat{S}_X^v[k]$ is one estimated SCF coefficient [7]. Equation (1) corrects the 2021 formula, which omitted frequency normalization and conflated a candidate-list index with physical center frequency.

The spectrum-sensor design searches and structures these SCF estimates. A scheduling algorithm manages requested symbol-rate candidates. Each request is processed by a streaming-SCA instance, then evaluated with the binary test

$$\left|\widehat{S}_X^v[k]\right|^2 \underset{H_0}{\overset{H_1}{\gtrless}} \beta\left(\frac{1}{N}\sum_{n=0}^{N-1}|X[n]|^2\right)^2. \tag{2}$$

The scalar β sets the operating tradeoff between the symbol-rate detection proportion P_d and the false-alarm proportion P_{fa} [7]. Valid results are stored for a prospective real-time spectrum-management algorithm.

The 2021 study generated a receiver operating characteristic (ROC) curve for the reported two-signal configuration in Table I. The repository does not contain the raw ROC observations, and its surviving simulation script uses different signal settings, so the numerical curve cannot be independently reproduced from the archived files.

Youden's statistic is a scalar, not an ROC coordinate [8]:

$$J = P_d + (1 - P_{fa}) - 1 = P_d - P_{fa}. \tag{3}$$

The archived script correctly maximizes $P_d - P_{fa}$ after averaging repeated false-alarm values, but it then uses that averaged-

TABLE I
REPORTED TWO-SIGNAL ROC CONFIGURATION

Signal	N	f_s (MHz)	R_s (MBd)	f_c (MHz)	M	r	Nominal E_b/N_0 (dB)
0	2^{12}	1.0	0.1	0.1	32	0.2	3.0
1	2^{12}	1.0	0.2	0.2	32	0.2	3.0

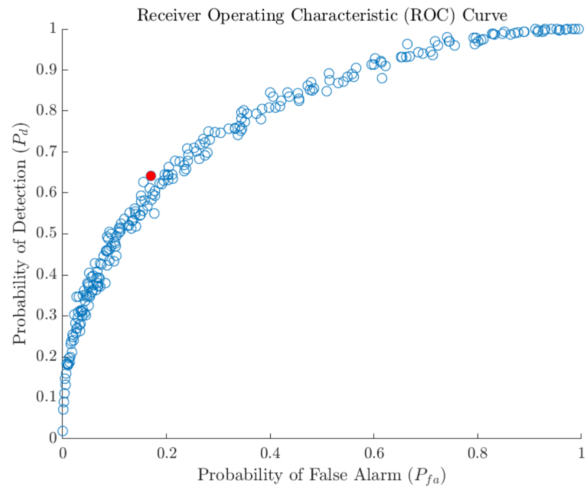

Fig. 6. Archived ROC curve based on 300 trials per value of β. The red point is approximately $(P_{fa}, P_d) = (0.17000, 0.64167)$, giving $J = 0.47167$. No confidence interval was calculated.

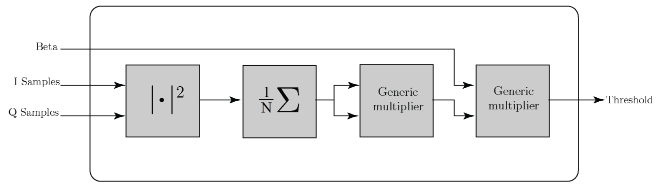

Fig. 7. Block diagram of the threshold module.

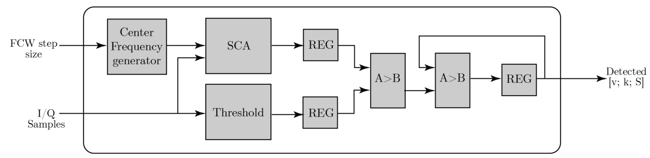

Fig. 8. Block diagram of the center-frequency detector module.

array index directly on the original β array. Consequently, the reported $\beta = 1.23 \times 10^{-4}$ is not reliably linked to the red point and is retained below only as the design value reported in 2021, not as a verified optimum. Maximizing J also gives equal weight to sensitivity and specificity; it does not uniquely minimize false alarms.

II. Digital Design

This section describes the digital-design additions proposed for autonomous spectrum sensing. A separately retained, mixed-date archive contains substantial RTL, behavioral traces, and a February 2022 hardware-manager capture. It is not an exact, reproducible 2021 revision: components name different target devices, and no complete top-level build, constraints, implementation reports, or power measurements were retained. The archive documents development activity but does not validate hardware performance.

A. Threshold

The threshold module receives in-phase and quadrature samples from an analog-to-digital converter. As shown in Fig. 7, samples are serially magnitude-squared, accumulated, right-shifted, and squared. The 2021 paper reported β as fixed before synthesis; the later retained top-level RTL instead exposes it as a runtime input.

B. Center-Frequency Detector

The 2021 block description placed the binary test in the center-frequency detector (CFD). In the later retained RTL, the threshold module computes the right-hand side of (2), and a symbol-rate preprocessor applies the comparison before candidates reach the CFD. The CFD itself has no threshold input: for each candidate v, it sweeps center-frequency FCWs at the configured step size and returns the largest statistic $\ell = |\widehat{S}_X^v[k]|^2$ with its k. Figure 8 therefore represents the published, earlier topology rather than the later snapshot.

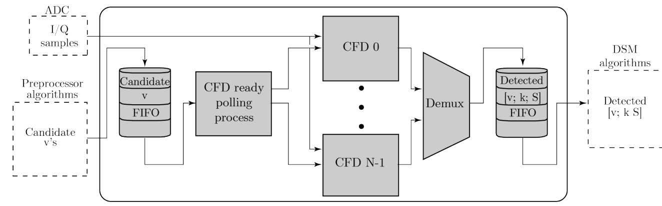

Fig. 9. Block diagram of the spectrum-sensor module.

TABLE II
Reported Symbol-Rate Detection Performance

Experiment	Nominal E_b/N_0 (dB)	r	P_d	P_{fa}
Baseline	15.0	0.35	1.0	0.0013
1	12.0	0.35	1.0	0.0284
2	9.0	0.35	1.0	0.0584
3	6.0	0.35	1.0	0.0854
4	3.0	0.35	0.9967	0.1146
5	0.0	0.35	0.89	0.1213
6	15.0	0.30	1.0	0.0108
7	15.0	0.25	1.0	0.0108
8	15.0	0.20	1.0	0.0150

C. Spectrum Sensor

The published spectrum-sensor design stores requested v values in an input FIFO and assigns them to available CFD instances. Historical simulation imagery retains the corresponding FIFO and scheduler signals, but their source is absent. The later snapshot instead feeds candidates directly to a 2022 priority-based distributor; its top-level flow-control process is commented out, and its result commutator relies on staggered CFD latency to avoid simultaneous outputs. Lossless arbitration was not demonstrably verified. Parallel result FIFOs store (v, k, ℓ) tuples for a prospective spectrum-management process. Figure 9 illustrates the intended, earlier design.

III. Results

The archived study averaged 300 trials for each scenario in Table II. The table preserves the published point estimates, but their labels require two qualifications. First, the scoring code tests symbol-rate membership only; it does not test whether the selected center frequency is correct. Thus, P_d is the detected fraction of true symbol-rate candidates and P_{fa} is the false-positive fraction of unoccupied symbol-rate candidates, rather than a whole-system probability. Second, the transmitter peak-normalizes each waveform and calls the noise routine without measuring its average input power; it also adds noise independently to each signal before summing them. The listed E_b/N_0 values are therefore nominal routine inputs, not verified received ratios. New quantitative conclusions require a corrected, seeded rerun.

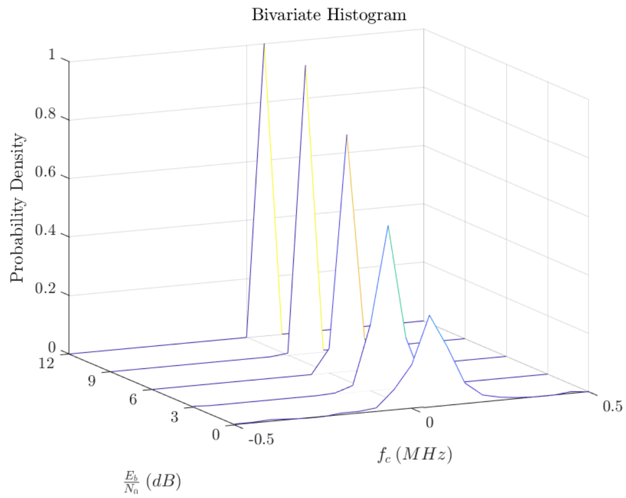

Fig. 10. Archived 300-record center-frequency rows for nominal $E_b/N_0 = 12, 9, 6, 3,$ and 0 dB. Height is per-bin relative frequency, not density. The plot omits one zero-valued record at 3 dB and 33 at 0 dB; dispersion among returned estimates increases as nominal E_b/N_0 falls.

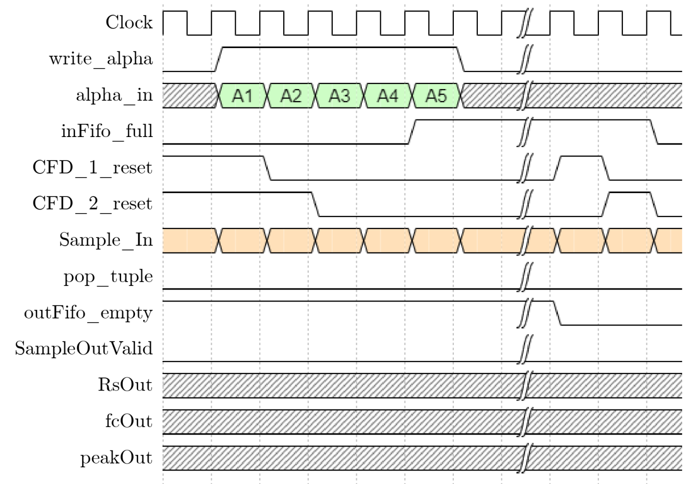

Fig. 11. Timing diagram for pushing v candidates to the input FIFO.

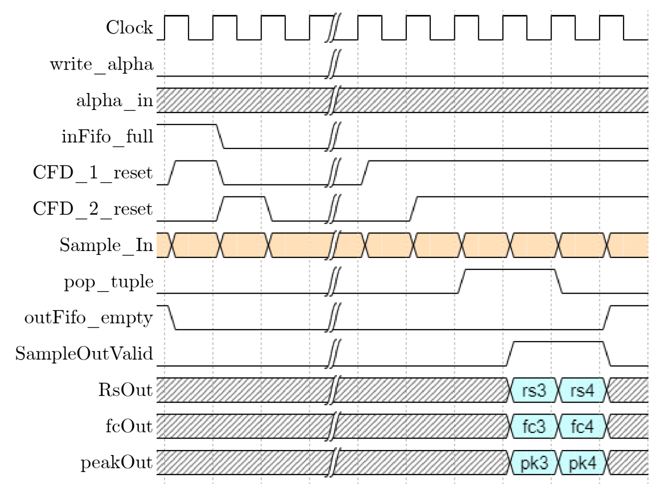

Fig. 12. Published output timing example. Later retained FIFO logic rejects a new write while full rather than discarding the oldest entry; source for the illustrated earlier behavior was not retained.

A. Characterization

At the reported design value $\beta = 1.23 \times 10^{-4}$, the symbol-rate detection proportion was 0.89 in the lowest nominal E_b/N_0 scenario and at least 0.9967 in the other eight scenarios. The reported false-alarm proportion ranged from 0.0013 to 0.1213. Because the threshold-to-ROC mapping is not recoverable and no confidence intervals or independent validation set were retained, these values neither verify an optimal threshold nor confirm ROC accuracy.

The archived scoring code explicitly counts a result as detected when its symbol-rate candidate is correct, even if its center-frequency candidate is inaccurate. A retained plotting script maps the five 300-record center-frequency files to nominal E_b/N_0 values of 12, 9, 6, 3, and 0 dB. It counts exact stored FCW values in nominal 0.05-MHz bins and divides by 300, so the ordinate is per-bin relative frequency, not probability density. Literal zeros are not one of those FCW bins: one is omitted at 3 dB and 33 at 0 dB, leaving plotted masses of 0.9967 and 0.89. Their equality to the corresponding P_d values strongly suggests that zero records denote misses, although the generator and acceptance rule are absent. The evidence supports increased dispersion among returned estimates as nominal E_b/N_0 decreases, but not calibrated center-frequency accuracy.

B. Timing

To illustrate the published scheduling scheme, consider two CFDs and input/output FIFOs of depth two. In Fig. 11, the upstream system supplies v candidates. The scheduler assigns A_1 and A_2 to the two CFDs. Candidates A_3 and A_4 wait until a CFD is available; A_5 is not accepted because the input FIFO is full, as indicated by `inFifo_full`. After processing, the results for A_1 and A_2 enter the result FIFO, allowing A_3 and A_4 to be assigned. The source for this earlier scheduler was not retained, so the timing diagram is illustrative rather than reproducible.

The published output example (Fig. 12) states that arriving A_3 and A_4 results replace the oldest entries when the result FIFO is full. The later retained `SimpleFifo` does the opposite: absent a simultaneous read, it rejects a write while full and preserves queued entries. Because the source underlying the timing figure is absent, the asserted overwrite-on-full behavior cannot be verified and is contradicted by the later implementation.

IV. Conclusion

This work proposed an FPGA-oriented spectrum-sensor architecture built around the streaming-SCA, with candidate scheduling, threshold detection, and result buffering. The archived Monte Carlo outputs characterize symbol-rate detections for simulated co-channel QAM–SRRC signals. Mixed-date RTL and a later hardware-manager capture document subsequent development, while the timing diagrams describe an earlier FIFO topology whose complete source was not retained. These records do not validate center-frequency accuracy, a fully blind search, implementation resource use, timing performance, power, end-to-end FPGA operation, or operation over a general RF channel.

Future work should first restore an executable and versioned simulation, normalize average signal power, add channel noise once after combining signals, define a candidate grid independently of ground truth, retain random seeds and raw observations, and rerun the ROC analysis with a correct mapping from each averaged operating point to β. Symbol-rate and center-frequency errors should then be scored separately with uncertainty intervals. Only after that recalibration should the detector be assessed on additional signal classes. Hardware claims require an exact, buildable RTL revision with its constraints and device configuration, implementation resource and timing reports, and power measurements.

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REFERENCES

[1] L. Plice, A. Dono Perez, and S. West, "HelioSwarm: Swarm mission design in high altitude orbit for heliophysics," in *Proc. 2019 AAS/AIAA Astrodynamics Specialist Conf.*, Portland, ME, USA, Aug. 2019, paper AAS 19-831. [Online]. Available: https://ntrs.nasa.gov/citations/20190029108

[2] H. Sanchez, D. McIntosh, H. Cannon, C. Pires, J. Sullivan, S. D'Amico, and B. O'Connor, "Starling1: Swarm technology demonstration," in *Proc. 32nd Annu. AIAA/USU Conf. Small Satellites*, Logan, UT, USA, Aug. 2018, paper SSC18-VII-08. [Online]. Available: https://digitalcommons.usu.edu/smallsat/2018/all2018/299/

[3] G. Giambene, S. Kota, and P. Pillai, "Satellite-5G integration: A network perspective," *IEEE Network*, vol. 32, no. 5, pp. 25–31, Sep. 2018. [Online]. Available: https://doi.org/10.1109/MNET.2018.1800037

[4] H. N. Cannon, H. S. Sanchez, and D. M. McIntosh, "Starling1—mission technologies overview," NASA Ames Research Center, Moffett Field, CA, USA, Presentation ARC-E-DAA-TN59780, Aug. 2018. [Online]. Available: https://ntrs.nasa.gov/citations/20180007374

[5] E. Kulu. (2021, Apr.) Total nanosatellites and CubeSats launched. Data through Apr. 4, 2021. [Online]. Available: https://www.nanosats.eu/img/fig/Nanosats_total_2021-04-04.pdf

[6] M. V. Koch and J. A. Downey, "Interference mitigation using cyclic autocorrelation and multi-objective optimization," NASA Glenn Research Center, Cleveland, OH, USA, Tech. Rep. NASA/TM—2019-220226, Jul. 2019. [Online]. Available: https://ntrs.nasa.gov/citations/20190027051

[7] D. J. Gormley, "A low-memory spectral-correlation analyzer for digital QAM-SRRC waveforms," Master's thesis, Cleveland State University, Cleveland, OH, USA, May 2021. [Online]. Available: https://rave.ohiolink.edu/etdc/view?acc_num=csu1622636550863441

[8] W. J. Youden, "Index for rating diagnostic tests," *Cancer*, vol. 3, no. 1, pp. 32–35, 1950. [Online]. Available: https://doi.org/10.1002/1097-0142(1950)3:1⟨32::AID-CNCR2820030106⟩3.0.CO;2-3